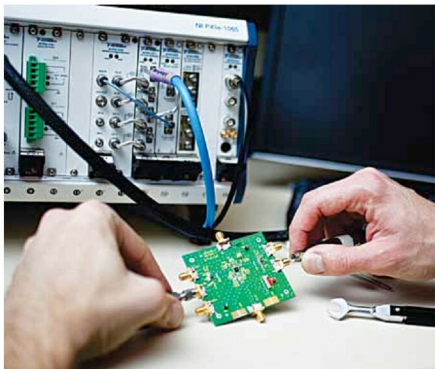


T&M Trends For RF Designers

The article looks at the emerging trends and applications in the RF design space which are driving the development of cutting-edge test and measurement tools



RF designers require custom measurements of the device under test

VANDANA SHARMA

"A lack of precision is dangerous when the margin of error is small"

— Donald Rumsfeld,
the American Secretary of Defense

These words hold true for any area of operation where a minuscule error can lead to critical repercussions. RF design testing is one such field where lack of accuracy can lead to the failure of a design project.

RF designers in fields such as telecom, aerospace and defence focus on building cutting-edge technologies and applications of the future. The test and measurement (T&M) industry is keeping up with these developments.

Recent trends

Demand for compact and sophisticated equipment. The latest trends in T&M equipment for RF design include instruments with higher frequency and higher instantaneous bandwidths in a compact PCI eXtensions for Instrumentation (PXI) form factor.

Abhay Samant, senior RF group manager, NI India R&D, informs, "In the recent months, we have seen PXI RF signal analysers with up to 14GHz frequencies, 50MHz instantaneous bandwidth and a dynamic range of 80 dB and higher. With very good interactive software support, designers can use these instruments on desktop for prototype design testing and early validation." Engineers can also use

very powerful instrument drivers for these instruments in LabVIEW, C, CVI or .Net to make custom measurements.

Equipment for the next-generation wireless standards. In the next-generation wireless communications standards, we see continued trends of using more spatial streams, wider channel bandwidths and higher-order modulation types to increase data throughput.

"As more sophisticated wireless standards evolve with more complex modulation/access techniques to address the efficient use of scarce resources like RF spectrum and power, demand for T&M equipment for these standards continues to increase. Apart from this, high-speed multimedia standards like wireless HDMI are also driving the need for advanced RF T&M equipment," says Vishal Gupta, RF applications consultant, Agilent Technologies.

Samant illustrates, "With IEEE 802.11ac, this trend has resulted in the use of 8×8 multiple-input and multiple-output (MIMO), as much as 160MHz bandwidth and 256-quadrature amplitude modulation. With LTE Advanced, support for 8×8 MIMO configurations and the implementation of carrier aggregation to support as much as 100MHz channel bandwidth are also seen."

However, while the next-generation wireless standards provide obvious benefits to consumers in the form of faster data transfer rates, design and test of IEEE 802.11ac and LTE Advanced radios pose significant challenges.

Tarun Gupta, business development manager, defense and telecom, NI India, explains, "From building transceivers capable of handling a wider bandwidth to packing more